CAPSTONE | PROFESSIONAL REVIEW II

Intelligent Traffic Management and Emergency Response Platform

A simulation-validated system for adaptive traffic control, incident awareness, and emergency priority on the Vijayawada corridor.

SUMO + TraCI | SAFE SIGNAL CONTROL | EMERGENCY ROUTING

PRESENTED BY (STUDENT INVESTIGATORS):

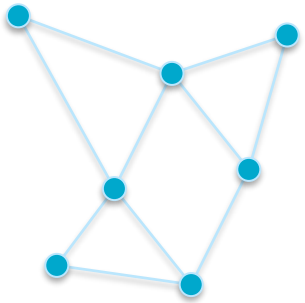
B. Prakash Raj (23BCE9434) | M. Venkat Komal (23BCE9377) | D. Preethi Darahasa (23BCE8318) | K. Vivek (23BCE9396)

Guide: N. Purnachand | SCOPE | VIT-AP University • REVIEW II | 2026

JOURNEY SINCE REVIEW I

From Planned System to Measurable Working Modules

The implementation now features an arterial corridor model, formal safety logic, emergency routing, and a trainable anomaly pipeline.



M0–M1

Foundation & Live Corridor

Docker, event contracts, Avro schemas, and Vijayawada SUMO corridor.

M2

Safety Kernel

Formal signal interlocks, conflict lockout, clearance, adversarial verification.

M3–M5

Adaptive Control

A-CA-MP policy, pressure-based cycle supervision, Webster benchmarks.

M6

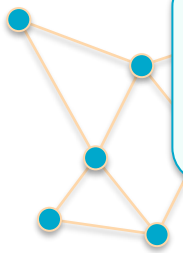
Anomaly Pipeline

Spatiotemporal graph detection, conformal calibration, human-in-the-loop gate.

M7

Emergency Response

Hospital-aware dispatch, dynamic routing, time-boxed green corridor.



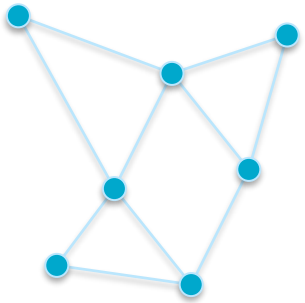
REVIEW II EVIDENCE PRINCIPLE:

We report measured gains where they are proven, and explicitly state the remaining M6 model-accuracy and live-integration gaps.

CONTEXT & MOTIVATION

Why One Connected Response Is Needed

Fixed plans, late reporting, and isolated emergency handling create avoidable delay and systemic risk.



01

Fixed Timers

Pre-set schedules cannot react to changing queues, sudden surges, or blocked lanes.

02

Late Incident Awareness

Road blockages propagate congestion upstream before operators have reliable evidence.

03

Emergency Delay

Ambulances face the same static signal sequence and queues as general traffic.

04

Fragmented Control

Traffic signal optimization, incident detection, and priority operate in separate silos.



OUR ARCHITECTURAL RESPONSE:

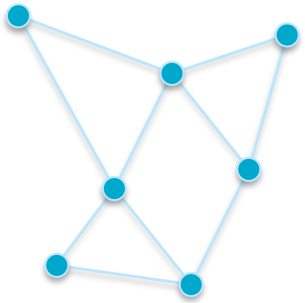
Sense traffic state → Decide safely → Coordinate emergency movement → Preserve an audit trail

The controller is deliberately constrained: every signal request is validated by a formal safety boundary before reaching SUMO.

SYSTEM ARCHITECTURE

A Closed-Loop System with Safety at the Command Boundary

One normalized event vocabulary lets SUMO, human operators, and future roadside sources feed the same decision flow.



01

SENSE

SUMO vehicle positions, loop detectors, operator inputs.

02

ANALYSE

Edge aggregation, spatiotemporal graph anomaly scores.

03

DECIDE

A-CA-MP adaptive signal policies, dynamic routing.

04

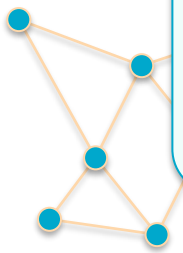
ARBITRATE

Safety kernel: min green, clearance, conflict lockout.

05

ACT

TraCI commands, operator UI fanout via WebSockets.



COMMON EVENT CONTRACT & REDIS BUS SPECIFICATION:

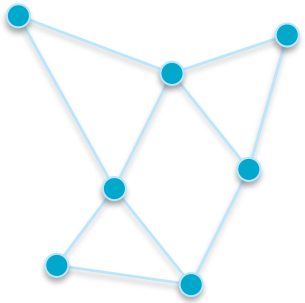
Vehicle positions | Signal changes / faults | Incident candidates / confirmed / cleared | Emergency commands

Strict Avro schemas enforced across all producers. Phase 1 operates over Redis Streams, architected for eventual Kafka scale.

SIMULATION TESTBED

The Vijayawada MG Road Corridor Is the Shared Evaluation Ground

The same high-fidelity model underpins controller, incident, and emergency-response experiments.



7

Signalized Junctions

+ 9 priority junctions across MG Road (Benz Circle to PNBS).

73

Controlled Links

Modelled movements feeding M6 spatiotemporal graph features.

103

Medical Facilities

Real hospital coordinates available for emergency destination routing.

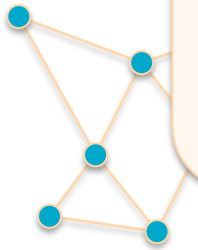
1 Hz

Live Map Updates

Tick-level state synchronization published between TraCI and UI.

RIGOROUS EVALUATION METHODOLOGY & PRINCIPLES:

- Held-Out Seed Consistency: Controller comparisons hold demand and random seeds strictly constant.
- Heterogeneous Traffic Mix: Calibrated distribution of cars, two-wheelers, auto-rickshaws, and buses.
- Real Geographic Topology: True OpenStreetMap road coordinates mapped into SUMO nodes and multi-phase signals.



A-CA-MP Makes Local Pressure Decisions Inside a Safe Cycle Envelope

The controller remains adaptive; the supervisor ensures each demanded phase is served before the bounded cycle closes.

CONTROL LOGIC FLOW

Observe Queues & Pressure
Calculates real-time occupancy and directional approach pressure.

Select Highest-Pressure Phase
Greedily picks phase maximizing clearing flow.

Arbiter Verification
Enforces min green, clearance (3s yellow+1s red), conflict lockout, max wait.

Cycle Supervisor Ceiling
Prevents side-street starvation under heavy demand.

-19.9%

Light-Demand Delay Reduction
vs. Tuned Webster baseline | 10/10 held-out evaluation seeds.

-11.4%

Medium-Demand Delay Reduction
vs. Tuned Webster baseline | Significant gate benchmark result.

-11.6%

Heavy-Demand Delay Reduction
vs. Tuned Webster baseline | Zero safety interlock violations.

Why the supervisor matters: Activates in 0.6–1.0% of phase decisions, yet eliminates heavy-demand violations.

The Detection Pipeline Is Built End-to-End; Model Accuracy Is the Current Gap

We separate implementation completeness from validation completeness and report proof-of-training metrics honestly.

01
FEATURES
73 links x 10 traffic features per tick (occupancy, speed, flow).

02
GRAPH MODEL
Temporal graph autoencoder scores unusual link reconstruction errors.

03
CALIBRATE
Conformal threshold + persistence and cooldown windows.

04
HUMAN GATE
ADR-006: Operator confirms or dismisses candidate via UI.

05
AUDIT
State machine with tamper-evident SHA-256 hash chains.

0.50 Recall
Current (Target: ≥ 0.90)
Signal detected; undertrained on edge cases.

40 s Latency
Current (Target: < 30 s)
Injection to candidate threshold crossing.

105+ Tests
Dedicated Automated Tests
Covers calibration, fusion, event contracts.

30 Runs
Next Step Evidence
Record normal traffic; train for 50–80 epochs.

Green-Corridor Coordination Reduces Emergency Transit Delay

Instead of each signal reacting in isolated panic, the entire route is reserved and released dynamically as the vehicle progresses.

3.5%

EV Travel Time Saved

vs. reactive preemption baseline across arterial runs.

25.3%

Signal Red Time Saved

Cumulative signal waiting duration eliminated for ambulances.

31.5%

Fewer Full EV Stops

Drastic reduction in emergency vehicle stop-and-go maneuvers.

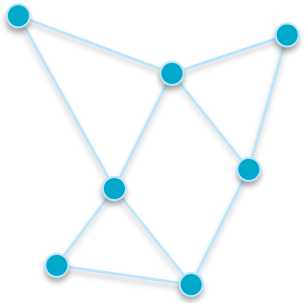
INTELLIGENT DISPATCH & GREEN CORRIDOR RESERVATION:

- Hospital Destination Matching: Hospital destinations are mapped from real OSM data; router picks fastest facility via live traffic cost.
- Time-Boxed Route Preemption: Signals along the route reserved just-in-time and immediately released upon EV clearance.
- Zero Side-Street Penalty: Delivers emergency priority with no statistically meaningful congestion penalty to cross-streets.

INTEGRATION & DEMONSTRABLE SYSTEM

The Live Operator Flow Drives the Controlled Simulation

The web application is not a mock-up: it receives live tick events, displays candidates, and can affect the running SUMO scenario.



STEP 1

SUMO Corridor

Controlled by A-CA-MP adaptive policy in TraCI.

STEP 2

Redis Event Stream

Standardized Avro event pub/sub envelope messaging.

STEP 3

API Service

FastAPI backend handles ingest, replay, and validation.

STEP 4

WebSocket Console

Real-time operator console with live corridor state.

STEP 5

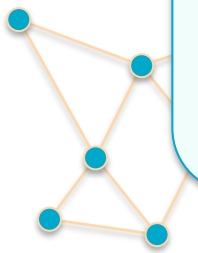
Approved Action

Operator confirm/reject commands fed back to corridor.

CURRENT VERIFIED LIVE CAPABILITIES:

- Live vehicle and signal state streaming at 1 Hz update frequencies.
- Candidate confirmation/rejection REST API with cryptographic audit chain logging.
- Operator-driven anomaly injection directly affecting vehicle behavior in SUMO.
- Dynamic emergency vehicle dispatching with live route re-calculation.

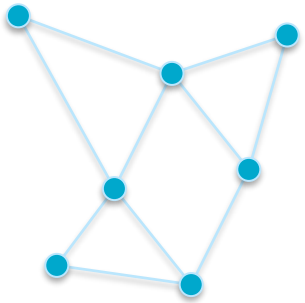
Note: Benchmarked M7 preemption is complete in simulation; wiring that engine into the live operator UI is an explicit remaining task.



VALIDATION DISCIPLINE

Safety Checks and Paired Benchmarks Gate Every Result

The project deliberately treats negative results and real-world caveats as scientific findings — not hidden exceptions.



PILLAR A

Safety First

9 property tests each generate 10,000 adversarial inputs via Hypothesis. Safety kernel rejects unsafe plans before reaching SUMO.

PILLAR B

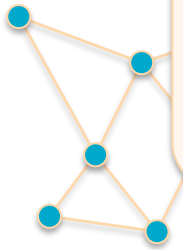
Fair Comparison

All controller arms share identical demand schedules, turn ratios, and random seeds. Backed by paired held-out seeds and 95% CIs.

PILLAR C

Reproducible Artefacts

Raw experiment logs, benchmarking scripts, Avro schemas, test suites, and regression gates preserved under version control.



WHAT WE DO NOT OVERCLAIM (HONEST ENGINEERING BOUNDARIES):

- M6 Graph Detector: Fully wired and tested in code, but currently undertrained due to normal-traffic data volume limits.
- Camera/Vision Stream (M8): Remains future scope; current validation relies on simulated loop detectors and graph autoencoders.
- Pedestrian & Mixed Flow: Simulation demand models multi-class vehicles, but pedestrian crossing behaviors remain simplified.

STATUS AND NEXT REVIEW PLAN

Closing Accuracy and Integration Gaps

A review-ready project makes its tested operational boundaries as clear as its achievements.

1

Validated Now

M0–M5 foundations, safe adaptive signal control (A-CA-MP), and M7 emergency dispatch/corridor preemption benchmarks.

2

Implemented, Needs Evidence

M6 spatiotemporal graph-detection pipeline; expand normal-data collection runs and train the shipped model.

3

Next Integration Milestone

Connect benchmarked M7 green-corridor preemption engine directly into the live operator console workflow.

4

Future Scope (M8+)

Camera-based YOLO vision detection stream, field-calibrated Vijayawada vehicle demand, pedestrian-actuated phases.

Thank You

Questions & Technical Discussion

We are prepared to demonstrate the live corridor simulation, explain the formal safety kernel & measurement gates, and discuss the M6 model training and deployment plan.

B. Prakash Raj (23BCE9434) | M. Venkat Komal (23BCE9377) | D. Preethi Darahasa (23BCE8318) | K. Vivek (23BCE9396)

Guide: N. Purnachand | SCOPE | VIT-AP University

CAPSTONE PROJECT • PROFESSIONAL REVIEW II • 2026